

Shedding the Bits: Pushing the Boundaries of Quantization with Minifloats on FPGAs

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Abstract—Post-training quantization (PTQ) is a powerful technique for model compression, reducing the numerical precision in neural networks without additional training overhead. Recent works have investigated adopting 8-bit floating-point formats (FP8) in the context of PTQ for model inference. However, floating-point formats smaller than 8 bits and their relative comparison in terms of accuracy-hardware cost with integers remains unexplored on FPGAs. In this work, we present minifloats, which are reduced-precision floating-point formats capable of further reducing the memory footprint, latency, and energy cost of a model while approaching full-precision model accuracy. We implement a custom FPGA-based multiply-accumulate operator library and explore the vast design space, comparing minifloat and integer representations across 3 to 8 bits for both weights and activations. We also examine the applicability of various integer-based quantization techniques to minifloats. Our experiments show that minifloats offer a promising alternative for emerging workloads such as vision transformers.

Index Terms—minifloats, multiply-accumulate, quantization

I. INTRODUCTION

With the increasing demand for deploying machine learning models on resource-constrained devices, post-training quantization (PTQ) [1]–[3] has been the prevailing choice for model compression, offering high model accuracy with reduced compute and memory usage. PTQ enables low-precision arithmetic without any model re-training and with minimal fine-tuning or calibration using only a small unlabeled dataset.

Multiply-accumulate (MAC) operations are ubiquitous in deep learning models. While GPUs with their high concentration of MACs remain the primary computing platform for these workloads, their support for different numerical precisions is inherently limited. In contrast, FPGAs offer unparalleled flexibility by supporting arbitrary formats at the bit level, presenting a unique advantage. Lately, several quantization works [4]–[6] have explored the idea of utilizing 8-bit floating-point (FP8) representations in lieu of integers. Integer formats are often preferred over floating-point alternatives due to their simpler hardware implementation. However, as the bit-width of floating-point formats decreases, so do their resource footprints. This leads us to hypothesize floating-point MACs with size and throughput akin to their integer counterparts.

For instance, prior research by Xilinx demonstrated a 7-bit floating-point representation that reduces resource consumption significantly while maintaining comparable or even

superior accuracy in comparison to the INT8 format [7]. This observation makes reduced-precision floating-point arithmetic on FPGAs a practical and valuable choice, especially for models like transformers [8], which frequently feature numerous outliers in their activation distribution [9].

Building upon these advancements, our study delves into the realm of *minifloat quantization* and explores the hardware efficiency of representations with fewer than 8 bits on FPGAs. Such minifloats with low bit-widths can exhibit better model performance compared to their integer counterparts, mostly due to their superior dynamic range with more values close to zero. Moreover, the flexibility to adjust both the number of exponent and mantissa bits allows us to mitigate the loss in model accuracy that often accompanies reduced precision.

Unfortunately, existing literature predominantly focuses on reduced-precision integer quantization, with limited exploration of minifloats in terms of accuracy-hardware tradeoffs. To address this gap, we undertake a comprehensive investigation of the design space encompassing minifloats and integer quantization. Our investigation spans precisions ranging from 3 to 8 bits, applied to both weights and activations across a set of deep learning vision workloads for the ImageNet classification task [10].

Furthermore, we implement custom bit-width MAC units for both minifloats and integers to analyze the impact of the number of exponent and mantissa bits within different minifloat formats on FPGAs. Finally, we investigate the effect of existing post-training optimization techniques, such as SmoothQuant [9], gradient-based learned rounding [2], and GPTQ [3] for minifloats. Our main contributions are:

- We propose a novel PTQ quantization framework for low-precision minifloats, ranging from 3 to 8 bits.
- We implement an operator library to realize custom bit-width integer and minifloat MACs on FPGAs.
- We thoroughly explore the accuracy-hardware trade-offs, providing an in-depth analysis of three prominent vision models – ResNet-18, MobileNetV2, and ViT-B-32 – based on our custom FPGA-based operator library.

Our experiments indicate that minifloat quantization typically outperforms integer quantization for bit-widths of four or more, both for weights and activations. However, when compared against our FPGA hardware cost model, integer quantization often retains its Pareto optimality due to its slightly smaller hardware footprint than minifloats at a given precision.

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